

Received February 7, 2019, accepted February 17, 2019, date of current version March 25, 2019.

Digital Object Identifier 10.1109/ACCESS.2019.2901753

Analysis of an Optimal Fault Attack on the LED-64 Lightweight Cryptosystem

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This work was supported in part by the National Natural Science Foundation of China under Grant 61571063, Grant 61501275, and Grant 61701141, in part by the Natural Science Foundation of Beijing under Grant 3182028, in part by the China Postdoctoral Science Foundation under Grant 2017M611357, in part by the Postdoctoral Science Foundation of Heilongjiang Province of China under Grant LBH-Z17045, in part by the Young Creative Talents Training Plan of General Universities of Heilongjiang Province of China under Grant UNPYSCT-2017152, in part by the Major Scientific Research Project of Zhejiang Lab under Grant 2018FD0ZX01, in part by the Alibaba-Zhejiang University Joint Institute of Frontier Technologies, and in part by the Open Fund of State Key Laboratory of Cryptology under Grant MMKFKT201805.

ABSTRACT This paper presents an optimal method for recovering the secret keys of the light encryption device (LED) by combining the impossible differential fault attack with the algebraic differential fault attack. The proposed optimal method effectively improves the performance of fault attacks. A fault attack model, named the redundant random nibble fault model (RRNFM), is proposed to simulate a multiple fault injection in a real-world environment. Using the optimal method and the RRNFM, the 64-bit secret key of LED can be recovered by injecting no more than six faults in 1300 experiments. We establish an equation of inverse proportionality between the number of faults injected and the remaining amount of the secret key. Using the equation, the number of fault injection can be accurately predicted, providing an effective way of evaluating fault attacks.

INDEX TERMS Impossible differential fault attack, LED, algebraic differential attack, redundant faults.

I. INTRODUCTION

Information security issues are very important in the military, industrial production, and people's daily lives. Information security experts want to protect confidentiality, integrity and availability of data throughout the system. However, wireless sensor networks (WSNs) placed in unattended outdoor environments are vulnerable to various attacks. Generally, terminal authentication and key management protocols are used to ensure the security of a WSN system. On the other hand, wireless sensor nodes need to directly store their respective secret key cryptographic algorithms. When the sensor nodes operate, they radiate noise-like signals. In this case, an adversary obtains the secret key by extracting electromagnetic leakage information, power consumption leakage data, and the running time of a node's chip [1]. Furthermore, if an attacker disguises the internal legitimate nodes in a WSN and sends

a series of attacks to other nodes, the entire network security will be severely damaged. These physical attacks are clearly more destructive than the traditional decryption analysis method. For instance, an attacker can inject a variety of faults into embedded devices [1]. The adversary can inject low-frequency AC voltage, perform clock-glitch attacks under the impact of heating [2], and inject a laser beam as well as an electromagnetic (EM) fault [1]. In 1996, Boneh *et al.* [18] successfully broke the RSA cryptosystem by injecting faults. Fault attacks pose a serious threat to the security of WSNs. Therefore, it is very important to investigate the security of the cipher algorithm against fault attacks.

Lightweight cryptosystems with the advantages of low energy consumption and small storage space have received increasing attention in the mobile payment and WSN security domains [19]. The LED (Light Encryption Device) block cipher proposed by Guo *et al.* [3] at CHES 2011 is compact in hardware and performs well in a software implementation. This lightweight encryption algorithm has become a research

The associate editor coordinating the review of this manuscript and approving it for publication was Jafar A. Alzubi.

hotspot since it was proposed. To date, many novel attack methods that estimate the security of the LED, such as multicollisions [4], the super-S-box [5] and the Even-Mansour scheme [6] have been introduced. Algebraic differential fault attack (ADFA) [14], [15], [17] and impossible differential fault attack (IDFA) [13], [16] are widely used for evaluating LED safety. To the best of our knowledge, no fusion method has been proposed for attacking LED. Considering a high level of safety, it is necessary to evaluate the security of the LED by the fusion method. In this paper, we propose a method by combining the IDFA with the ADFA on the LED-64 cryptosystem.

In 1997, Biham and Shamir [8] proposed a differential fault attack (DFA) for DES by combining a fault attack and differential cryptanalysis. DFA can significantly reduce the secret key space of a cryptosystem [7], [9], [23], [25]. However, the impossible differential attack is a special case of differential attacks, which was first introduced by Knudsen [10] and Biham *et al.* [11] to analyze DEAL and Skipjack, respectively. Impossible differential attack is the probability that its difference in a certain state of the cryptosystem is zero. The attacker excludes secret key candidates by constructing impossible differential pairs in both directions. For block ciphers, IDFA is very effective and can reduce the secret key space faster than any other algorithm. Impossible meet-in-the-middle fault attack (IMITMFA) is a kind of IDFA that uses the impossible conditions of meeting in the middle to eliminate the secret key space. For instance, Li *et al.* [12], [13] injected 48 random faults by using IDFA and then injected 44.2 random faults by using IMITMFA in theory.

Algebraic fault attack (AFA) is widely used in cryptosystems [14], [15], [20], [24]. Generally, the attacker builds a set of algebraic equations and fault attack equations to analyze the cryptosystem; then, these equations are put into an automatic math solver. AFA can recover the secret key through a few steps. P. Jovanovic injected a known location fault at the beginning of the 30th round. Erroneous secret keys can be largely eliminated by using algebraic derivation operations [14]. However, there has been no discussion of multiple fault attacks on the LED. X. Zhao injected a fault in a fixed location [15]. The researcher used algebraic equations and the CryptoMiniSat solver to recover the secret key in no more than three minutes. However, it takes thousands of variables to set up the equations. Therefore, if an attacker can accurately estimate the fault location, the attack time will be reduced.

IDFA is constrained by many conditions, such as impossible differential methods used by attackers, their computing power, etc. However, currently popular research methods tend to inject fewer faults and use more effective fault injection methods and practical fault models. At present, the random nibble fault injection and continuous byte-fault injection method are widely studied in LED cryptosystems [12], [13], [16]. However, in the actual fault attacks, it is difficult to determine the number of faults and the internal bit position.

The use of precision laser pulses to inject the faults can also be constrained by various conditions, such as environmental disturbances, the time of fault injection, the laser spot size, etc. During the actual attacks, the laser is often interfered with by factors such as the external temperature and physical vibration of the surrounding environment [21], [22]. As a result, the attacker cannot determine whether the performed attack is a single-byte random fault or a nibble random fault, which causes significant difficulties for practical analysis.

Considering the abovementioned factors, we propose an effective fault-attack model named redundant random nibble fault model (RRNFM). It is assumed that a random nibble fault is injected; the RRNFM randomly adds two nibble faults with unknown values and at unknown locations near the nibble fault at the same time. The two random faults are used to simulate redundant faults that may occur. We propose an optimal solution for solving the above problem by combining the IDFA and the ADFA. To predict the number of fault attacks, an inverse proportional function relationship of the remaining amount of the secret key is established. After injecting the first fault, the required number of attacks can be predicted based on this relationship. In this paper, we improve previous results; the number of faults is reduced to no more than 6, and the attack time is less than 8 minutes.

The rest of this paper is organized as follows. In Section 2, we give a brief description of LED and the notation. In Section 3, we propose the fault model and the impossible differential fault algebraic attack. In Section 4, the method proposed above is verified and analyzed mathematically, and validated experimentally. In the last section, we conclude this paper.

II. SPECIFICATION OF LED

A. THE STRUCTURE OF LED

LED is a lightweight SPN block cipher with an AES-like architecture designed by Guo *et al.* [3]. Fig. 1 shows a 64-bit secret key array. For a 64-bit secret key (resp., 128-bit secret key), the step size equals 8 (resp., step=12). Each step consists of four rounds of operations – AddConstants, SubCells, ShiftRows, and MixColumnsSerial – in a sequence. Between the steps, the operation AddRoundKey performs bitwise exclusive-or with the state. Its secret key schedule is very efficient. All bits are disturbed only by one step.

$$\begin{bmatrix} 4 & rc_5 \parallel rc_4 \parallel rc_3 & 0 & 0 \\ 5 & rc_2 \parallel rc_1 \parallel rc_0 & 0 & 0 \\ 2 & rc_5 \parallel rc_4 \parallel rc_3 & 0 & 0 \\ 3 & rc_2 \parallel rc_1 \parallel rc_0 & 0 & 0 \end{bmatrix}$$

- 1) AddConstants (AC) is an operation where the layer uses a bitwise exclusive-or with the input matrix. The adversary computes its corresponding value in tables 1 and 2.
- 2) SubCells (SB) is an operation, whereby a nonlinear bijection perturbation layer and the PRESENT S-box are used to replace the nibble in the array state.

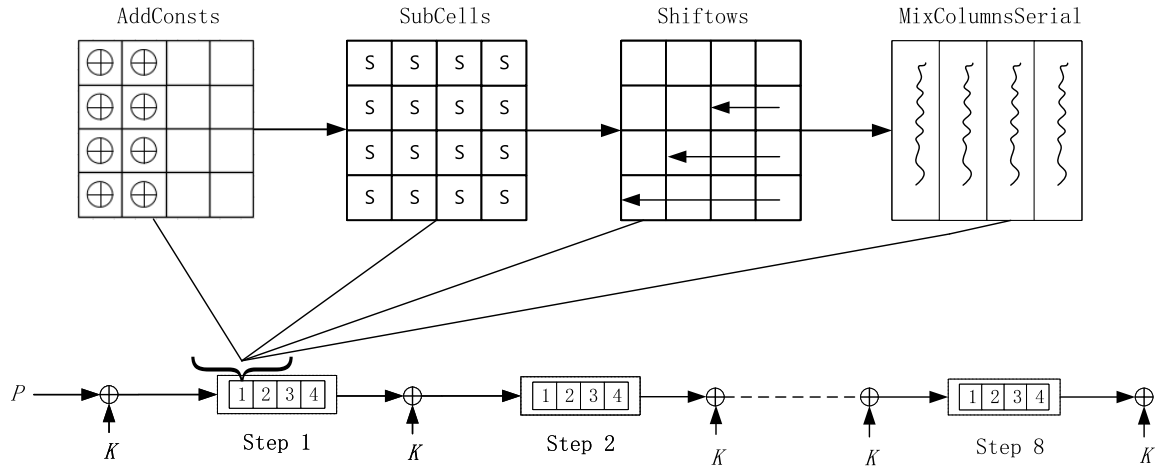


FIGURE 1. The structure of LED-64.

TABLE 1. Constant distribution 1.

Rounds	$rc_5 \ rc_4 \ rc_3$															
1-16	0	0	0	1	3	7	7	6	5	3	7	7	6	4	1	
17-31	3	7	6	5	2	5	3	6	4	0	0	1	2	5	3	7
32-48	6	4	0	1	3	6	5	3	6	5	2	4	1	2	4	0

TABLE 2. Constant distribution 2.

Rounds	$rc_2 \ rc_1 \ rc_0$															
1-16	1	3	7	7	6	5	3	7	7	6	4	1	3	7	6	
17-31	5	2	5	3	6	4	0	0	1	2	5	3	7	6	4	0
32-48	1	3	6	5	3	6	5	2	4	1	2	4	0	1	2	4

- 3) ShiftRows(SR) is an operation where for $n = 1, 2, 3 \dots 16$, the n -th position of the state matrix is cyclically shifted to the left by $(n-1)/4$ positions.
- 4) MixColumnsSerial (MC) is an operation that changes the value in the column state. The value of the state is updated by multiplying with matrix M .

$$M = \begin{pmatrix} 4 & 1 & 2 & 2 \\ 8 & 6 & 5 & 6 \\ B & E & A & 9 \\ 2 & 2 & F & B \end{pmatrix}$$

B. NOTATION

The following notation is used in the analysis of the 64-bit LED cryptosystem. The symbol $\oplus$ denotes bitwise exclusive-or, and $\|$ denotes bit connection. Let P be the plaintext. The right ciphertext is denoted by C , and the fault ciphertext is denoted by c^* . We denote the state of the r -th AC, SB, SR and MC by U_r^{AC} , V_r^{SB} , W_r^{SR} , and X_r^{MC} ; their differences are represented by ΔU_r^{AC} , ΔV_r^{SB} , ΔW_r^{SR} and ΔX_r^{MC} . Let $\Delta U_{r,\{1,2,3,4\}}^{AC}$ be the AC-state differential values of the 1st, 2nd, 3rd and 4th

columns in the r -th round. We denote the inverse operations of AddConstants, SubCells, ShiftRows and MixColumnsSerial by INVAC, INVSB, INVSr and INVMC.

III. THE OPTIMAL ATTACK

Although the subkey candidates can be excluded by the impossible differential fault path, this approach does not take into account the detailed calculation relationship between the rounds. If ADFA is applied separately, the impossible data will also be introduced into the calculations.

Therefore, this paper combines the ADFA with the IDFA to recover the secret key of the LED-64 cryptosystem. This method, named impossible differential fault algebraic attack (IDFAA), can perfectly combine the algebraic differential fault attack with the impossible differential fault attack, reducing the data complexity in the attack; its application is very effective in evaluating the security of the LED-64 cryptosystem.

A. THE FAULT MODEL AND ASSUMPTIONS

The fault model makes the following assumptions: first, the attacker has the ability to induce an arbitrary number of up to three-nibble faults in the 30th round, while the fault's locations and values are both unknown. Second, the attacker can specify the plaintext to obtain the corresponding ciphertext.

The adversary injects a random fault. However, due to the environmental factors, the adversary does not know how many faults have actually been injected. The RRNFM assumes that when a random fault is injected, two additional random faults are simultaneously introduced in the round. These two faults are used to simulate the actual redundant faults.

B. RRNFM

As chip size continues to shrink, fault injection techniques face new challenges. Fault injection is affected by many

factors such as laser spot diameter, environmental disturbance, and instrument parameters [21]. Therefore, during a fault attack, the fault often includes several random faults that the attackers themselves are unaware of. In this case, analysis of fault attacks using the single-byte or single-nibble fault attack model will encounter difficulties. In contrast, our RRNFM takes into account the actual situation presented above. Because of the random disturbance in the fault injection, the RRNFM considers the fault injection in the real-world by randomly generating two faults around the main attack point, while the fault's locations and values are both unknown. This method can effectively simulate fault injection in the real-world.

C. ATTACK IDEA OF IDFAA

IDFAA is the combination of impossible differential fault attacks and algebraic differential fault attacks. The classic ADFA can accurately calculate the secret key, but it requires a large amount of computation. Moreover, there are many redundant computations in ADFA. To reduce irrelevant data in the calculation, the IDFAA is proposed. This method includes three steps. First, according to the fault injection model (RRNFM), the faults are injected at any position in the 30th round of the LED cryptosystem. The fault difference in the 31st round S-box must exist at a certain column of a zero difference; then, the difference that does not satisfy the condition will be excluded. Second, the adversary uses the impossible differential path to classify the estimated ΔU_{32}^{AC} , and then substitutes ΔU_{32}^{AC} and ΔV_{32}^{SB} into the inverse differential S-box equation (1) to obtain set V_{32}^{SB} . Third, after repeated injection of faults, only one element in collection V_{32}^{SB} will eventually be left. If the number of elements in set V_{32}^{SB} is not one, the adversary continues to inject faults until the number of V_{32}^{SB} becomes one. The differential pairs of the inverse S-box are defined by

$$S^{-1}(V_r^{SB}) \oplus S^{-1}(V_r^{SB} \oplus \Delta V_r^{SB}) = \Delta U_r^{AC} \quad (1)$$

D. SETTING UP ALGEBRAIC EQUATIONS

1) INVERSE S-BOX AND INVERSE DIFFERENTIAL S-BOX

We first establish the inverse S-box Boolean equation by converting inverse the S-box Truth Tables into Boolean expressions.

$$x_1 = y_1 + y_2 + y_3 + y_4 + y_1y_2y_4 + y_3y_4 + y_2y_3y_4 \quad (2)$$

$$x_2 = 1 + y_1 + y_1y_3 + y_2y_3 + y_1y_4 + y_2y_4 + y_1y_2y_4 + y_3y_4 + y_1y_3y_4 + y_2y_3y_4 \quad (3)$$

$$x_3 = y_1 + y_1y_2 + y_3 + y_1y_3 + y_4 + y_2y_4 + y_1y_2y_4 + y_1y_3y_4 + y_2y_3y_4 \quad (4)$$

$$x_4 = 1 + y_2 + y_1y_3 + y_4 \quad (5)$$

According to the above inverse S-box expressions, we can further obtain its Boolean differential equations.

$$\begin{aligned} \Delta x_1 &= \Delta y_1 + \Delta y_3 + y_2y_4\Delta y_1 + \Delta y_2 + y_1y_4\Delta y_2 + y_3y_4\Delta y_2 \\ &\quad + y_4\Delta y_1\Delta y_2 + y_4\Delta y_3 + y_2y_4\Delta y_3 + y_4\Delta y_2\Delta y_3 \\ &\quad + y_1y_2\Delta y_4 + y_3\Delta y_4 + y_2y_3\Delta y_4 + y_2\Delta y_1\Delta y_4 \\ &\quad + y_1\Delta y_2\Delta y_4 + y_3\Delta y_2\Delta y_4 + \Delta y_3\Delta y_4 + y_2\Delta y_3\Delta y_4 \\ &\quad + \Delta y_1\Delta y_2\Delta y_4 + \Delta y_2\Delta y_3\Delta y_4 \end{aligned} \quad (6)$$

$$\begin{aligned} \Delta x_2 &= \Delta y_1 + y_3\Delta y_1 + y_4\Delta y_1 + y_2y_4\Delta y_1 + y_3y_4\Delta y_1 + y_3\Delta y_2 \\ &\quad + y_4\Delta y_2 + y_1y_4\Delta y_2 + y_3y_4\Delta y_2 + y_4\Delta y_1\Delta y_2 + y_1\Delta y_3 \\ &\quad + y_2\Delta y_3 + y_4\Delta y_3 + y_1y_4\Delta y_3 + y_2y_4\Delta y_3 + \Delta y_1\Delta y_3 \\ &\quad + y_4\Delta y_1\Delta y_3 + \Delta y_2\Delta y_3 + y_4\Delta y_2\Delta y_3 + y_1\Delta y_4 + y_2\Delta y_4 \\ &\quad + y_1y_2\Delta y_4 + y_3\Delta y_4 + y_1y_3\Delta y_4 + y_2y_3\Delta y_4 + \Delta y_1\Delta y_4 \\ &\quad + y_2\Delta y_1\Delta y_4 + y_3\Delta y_1\Delta y_4 + \Delta y_2\Delta y_4 + y_1\Delta y_2\Delta y_4 \\ &\quad + y_3\Delta y_2\Delta y_4 + \Delta y_1\Delta y_2\Delta y_4 + \Delta y_3\Delta y_4 + y_1\Delta y_3\Delta y_4 \\ &\quad + y_2\Delta y_3\Delta y_4 + \Delta y_1\Delta y_3\Delta y_4 + \Delta y_2\Delta y_3\Delta y_4 \end{aligned} \quad (7)$$

$$\begin{aligned} \Delta x_3 &= \Delta y_1 + y_3\Delta y_1 + y_2\Delta y_1 + y_2y_4\Delta y_1 + y_3y_4\Delta y_1 + y_1\Delta y_2 \\ &\quad + y_4\Delta y_2 + y_1y_4\Delta y_2 + y_3y_4\Delta y_2 + \Delta y_1\Delta y_2 + y_4\Delta y_1\Delta y_2 \\ &\quad + \Delta y_3 + y_1\Delta y_3 + y_1y_4\Delta y_3 + y_2y_4\Delta y_3 + \Delta y_1\Delta y_3 \\ &\quad + y_4\Delta y_1\Delta y_3 + y_4\Delta y_2\Delta y_3 + \Delta y_4 + y_2\Delta y_4 + y_1y_2\Delta y_4 \\ &\quad + y_1y_3\Delta y_4 + y_2y_3\Delta y_4 + y_2\Delta y_1\Delta y_4 + y_3\Delta y_1\Delta y_4 \\ &\quad + \Delta y_2\Delta y_4 + y_1\Delta y_2\Delta y_4 + y_3\Delta y_2\Delta y_4 + \Delta y_1\Delta y_2\Delta y_4 \\ &\quad + y_1\Delta y_3\Delta y_4 + y_2\Delta y_2\Delta y_4 + \Delta y_1\Delta y_2\Delta y_4 + \Delta y_3\Delta y_4 \\ &\quad + y_1\Delta y_3\Delta y_4 + y_2\Delta y_3\Delta y_4 + \Delta y_1\Delta y_3\Delta y_4 + \Delta y_2\Delta y_3\Delta y_4 \end{aligned} \quad (8)$$

$$\begin{aligned} \Delta x_4 &= y_3\Delta y_1 + \Delta y_2 + y_1\Delta y_3 + \Delta y_1\Delta y_3 + \Delta y_4 \end{aligned} \quad (9)$$

where $\Delta y_1, \Delta y_2, \Delta y_3, \Delta y_4$ indicate the differential input of the inverse S-box, and $\Delta x_1, \Delta x_2, \Delta x_3, \Delta x_4$ denote its output.

2) ALGEBRAIC EQUATIONS OF MC AND MC^{-1}

The MixColumnsSerial operation is a multiplication by a constant matrix in $GF(2^4)$. The algebraic equations of MC are represented as follows:

$$\begin{pmatrix} 4 & 1 & 2 & 2 \\ 8 & 6 & 5 & 6 \\ B & E & A & 9 \\ 2 & 2 & F & B \end{pmatrix} \cdot \begin{pmatrix} X_1 & X_2 & X_3 & X_4 \\ X_5 & X_6 & X_7 & X_8 \\ X_9 & X_{10} & X_{11} & X_{12} \\ X_{13} & X_{14} & X_{15} & X_{16} \end{pmatrix} = \begin{pmatrix} Y_1 & Y_2 & Y_3 & Y_4 \\ Y_5 & Y_6 & Y_7 & Y_8 \\ Y_9 & Y_{10} & Y_{11} & Y_{12} \\ Y_{13} & Y_{14} & Y_{15} & Y_{16} \end{pmatrix} \quad (10)$$

$$\begin{aligned}
Y_{1+l} &= 4 \cdot X_{1+l} \oplus 1 \cdot X_{5+l} \oplus 2 \cdot X_{9+l} \oplus 2 \cdot X_{13+l} \\
Y_{5+l} &= 8 \cdot X_{1+l} \oplus 6 \cdot X_{5+l} \oplus 5 \cdot X_{9+l} \oplus 6 \cdot X_{13+l} \\
Y_{9+l} &= B \cdot X_{1+l} \oplus E \cdot X_{5+l} \oplus A \cdot X_{9+l} \oplus 9 \cdot X_{13+l} \\
Y_{13+l} &= 2 \cdot X_{1+l} \oplus 2 \cdot X_{5+l} \oplus F \cdot X_{9+l} \oplus B \cdot X_{13+l} \\
l &= 0 \dots 3. \quad (11)
\end{aligned}$$

The algebraic equations of MC^{-1} are represented as follows:

$$\begin{pmatrix} C & C & D & 4 \\ 3 & 8 & 4 & 5 \\ 7 & 6 & 2 & E \\ D & 9 & 9 & D \end{pmatrix} \cdot \begin{pmatrix} Y_1 & Y_2 & Y_3 & Y_4 \\ Y_5 & Y_6 & Y_7 & Y_8 \\ Y_9 & Y_{10} & Y_{11} & Y_{12} \\ Y_{13} & Y_{14} & Y_{15} & Y_{16} \end{pmatrix} = \begin{pmatrix} X_1 & X_2 & X_3 & X_4 \\ X_5 & X_6 & X_7 & X_8 \\ X_9 & X_{10} & X_{11} & X_{12} \\ X_{13} & X_{14} & X_{15} & X_{16} \end{pmatrix}, \quad (12)$$

$$\begin{aligned}
X_{1+l} &= C \cdot Y_{1+l} \oplus C \cdot Y_{5+l} \oplus D \cdot Y_{9+l} \oplus 4 \cdot Y_{13+l} \\
X_{5+l} &= 3 \cdot Y_{1+l} \oplus 8 \cdot Y_{5+l} \oplus 4 \cdot Y_{9+l} \oplus 5 \cdot Y_{13+l} \\
X_{9+l} &= 7 \cdot Y_{1+l} \oplus 6 \cdot Y_{5+l} \oplus 2 \cdot Y_{9+l} \oplus E \cdot Y_{13+l} \\
X_{13+l} &= D \cdot Y_{1+l} \oplus 9 \cdot Y_{5+l} \oplus 9 \cdot Y_{9+l} \oplus D \cdot Y_{13+l} \\
l &= 0 \dots 3. \quad (13)
\end{aligned}$$

3) ALGEBRAIC EQUATIONS OF SR AND SR^{-1}

The ShiftRows (resp., Inverse ShiftRows) transformation is performed by cyclically shifting a certain number of nibbles to the left (resp., right). The shifted number of nibbles in each row is determined according to the following table 3:

TABLE 3. The number of nibbles.

Row number	Right shift	Left shift
1	0	0
2	1	1
3	2	2
4	3	3

4) ATTACK PROCESS

According to the method proposed above, we assume that no more than three random nibble-faults are injected into the 30th round. As shown in Fig. 2, the faults may be induced on either U_{30}^{AC} , V_{30}^{SB} or W_{30}^{SR} , whereas their fault propagation paths are identical. The attacker can obtain the correct ciphertext C , the fault ciphertext C^* , and their difference:

$$\Delta C = C \oplus C^* \quad (14)$$

By the impossible differential path in Fig. 2, we can obtain the following expressions:

$$\Delta V_{31}^{SB} = \text{INVS}R(\text{INVMC}(\Delta U_{32}^{AC})), \quad (15)$$

$$V_{32}^{SB} = \text{INVS}R(\text{INVMC}(C \oplus K)) = C' \oplus K', \quad (16)$$

$$V_{32}^{SB} = \text{INVS}R(\text{INVMC}(C^* \oplus K)) = C'^* \oplus K' \quad (17)$$

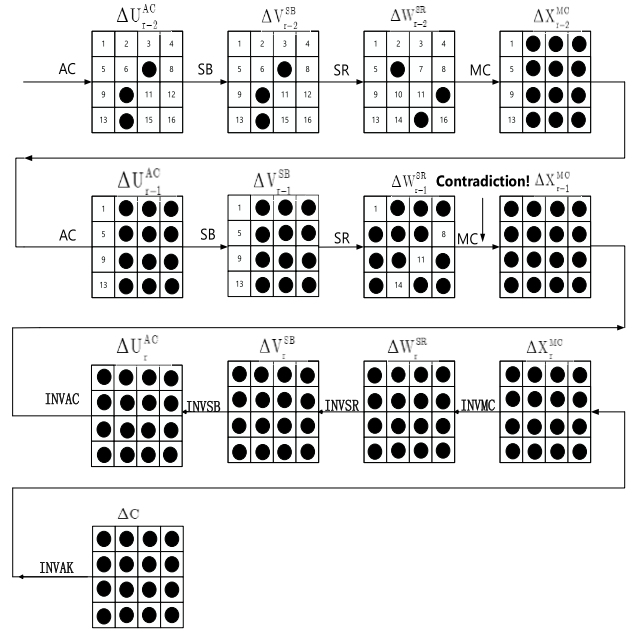


FIGURE 2. Impossible differential fault propagation path.

where

$$C' = \text{INVS}R(\text{INVMC}(C)), \quad (18)$$

$$C'^* = \text{INVS}R(\text{INVMC}(C^*)), \quad (19)$$

$$K' = \text{INVS}R(\text{INVMC}(K)), \quad (20)$$

and

$$\Delta U_r^{AC} = U_r^{AC} \oplus U_r^{AC*}. \quad (21)$$

According to preconditions, we can observe the impossible differential propagation path, in which there is at least one column of zeros in ΔV_{31}^{SB} . The adversary can divide ΔU_{32}^{AC} into four sets. The adversary substitutes ΔV_{32}^{SB} into the inverse differential S-box algebraic expression (1); then, the estimated ΔU_{32}^{AC} is obtained by the constraint of $\Delta V_{31,\{1\}}^{SB} = 0$. According to the same method, the estimated other sets of ΔU_{32}^{AC} are obtained by the constraints of $\Delta V_{31,\{2\}}^{SB} = 0$, $\Delta V_{31,\{3\}}^{SB} = 0$ and $\Delta V_{31,\{4\}}^{SB} = 0$. Substituting ΔV_{31}^{SB} and the estimated ΔU_{32}^{AC} into the differential equation (15), we can determine set V_{32}^{SB} . Following the method mentioned above, the adversary can take the intersection of V_{32}^{SB} until there is only one element in set V_{32}^{SB} . Finally, the secret key can be solved for by the following formula:

$$K = \text{MC}(\text{SR}(V_{32}^{SB})) \oplus C. \quad (22)$$

IV. EXPERIMENTAL ANALYSIS AND RESULTS

A. EXPERIMENTAL ANALYSIS

Lemma: The number of the estimated secret keys in the key set is not a constant, and it will vary according to the locations and values of faults. The locations and values of faults are random; therefore, it is difficult to accurately obtain

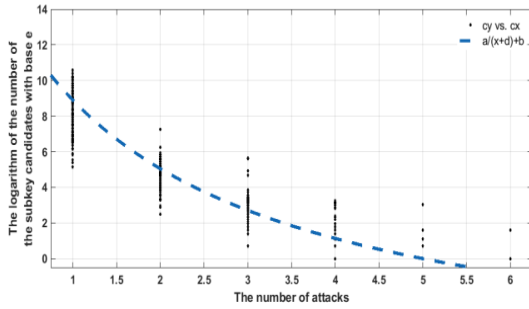


FIGURE 3. The function curve of the number of attacks and the number of remaining keys.

the number of initial candidate secret keys. The more attacks there are, the lower the remaining number of keys are in the secret key sets. Therefore, there is an inversely proportional relationship between the number of fault injection and the remaining number of secret keys. We establish an equation of inverse proportionality as follows:

$$y = \frac{a}{x+d} + b \quad (23)$$

where y is the logarithm of the number of residual secret keys with base e after each attack, x is the number of attacks, and a, b, d represent the initial constants.

According to the method proposed above, we performed 1,300 independent trials. After complex experimental data fitting operations, we get formula (24). Meanwhile, we use a 95% confidence level to find the confidence interval for $a \in [49.04, 49.19]$, $b \in [-6.921, -6.905]$ and $d \in [2.102, 2.108]$. Based on the formula (24), the relationship between the number of attacks and remaining subkeys after each attack is shown in Fig. 3. The equation can be expressed as

$$y = \frac{49.12}{x+2.105} - 6.913 \quad (24)$$

We evaluate the accuracy of (24) by SSE (Sum of Squared Error), R-squared and RMSE (Root Mean Squared Error), which are 3.453, 0.9999 and 0.02303, respectively.

Set V_{32}^{SB} is constantly intersected until there is only one element in the collection. Therefore, we can obtain the number of theoretical attacks. The range of x is (4.99,5.01) at a 95% confidence level.

Fig. 4 shows the proportional relationship of the number of attacks in 1300 trials. It is observed from Fig.4 that the total number of attacks is between 4 and 6.

Latency is the time interval between the first fault injection and the recovery of the subkey in the proposed software simulation. Fig.5 shows the proportion of latency in the 1300 trials. In 1300 trials, 92% of attacks can recover the subkeys within 8 minutes.

B. EXPERIMENTAL RESULTS

The attack experiments have been performed on a PC with a Core (TM) i3 CPU with 4GB of RAM, using the

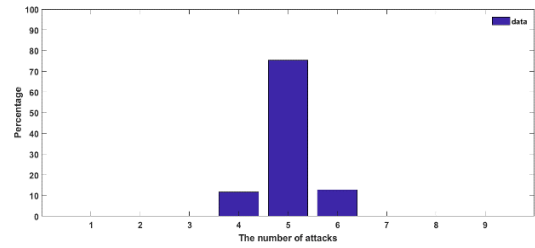


FIGURE 4. Distribution of the number of attacks.

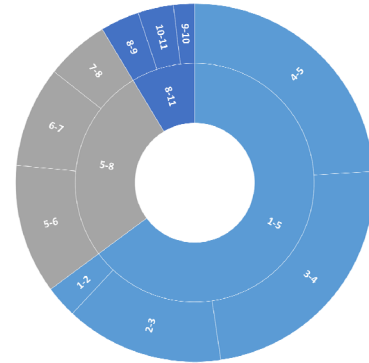


FIGURE 5. Distribution of Latency in IDFAA on LED-64.

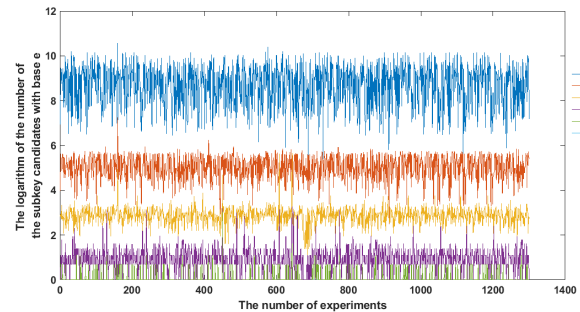


FIGURE 6. Intersection of attack times and subkey candidates in 1300 experiments.

TABLE 4. Experimental results for fault attacks on the LED-64.

Reference	Framework	Method	Position	Round	Value	Timing	Number of faults	Implemented tool	Experimental platform
5	single-key	DFA	unknown	R=29	1 nibble	7 m	1	C++	2G RAM
15	single-key	ADFA	unknown	R=30	1 byte	1 h	1	CryptoMiniSat	4G RAM
14	single-key	AFA	known	R=30	1 nibble	3.28-25.23 h	1	Minisat 2.2	8-core 50G RAM
12	single-key	IDFA	unknown	R=29	1 nibble	30-60 s	48	Java	32G RAM
13	single-key	IMITMFA	unknown	R=28	1 nibble	10-15 s	44.2	Java	32-core 64G RAM
this paper	single-key	IDFAA	unknown	R=30	1-3 nibble	1-11 m	4-6	Matlab	i3 4G RAM

MATLAB language. Fig. 6 illustrates the intersections of set V_{32}^{SB} , where the x-coordinate represents the number of experiments and the y-coordinate represents the subkey candidates. It is shown that the subkey can be recovered with a maximum of 6 attacks. Each of the colored lines represents the number of remaining subkeys. Table 4 compares our new results based on the RRNFM to the previously reported results for the LED-64 cryptosystem. The RRNFM proposed

in this paper is more effective than the random nibbles method in [5], [12], and [14] and double-nibble models in [15] in practice. The required number of faults for our method is less than that of [12] and [13], and its performance is better than that of methods in [5], [14], and [15] in terms of time complexity. With respect to the number of faults and time complexity, the method proposed in this paper outperforms the methods reported in [5] and [12]–[15].

V. CONCLUSION

A novel IDFAA method is applied to the LED-64 cryptosystem for the first time. To the best of our knowledge, little research combining the IDFA with the ADFA on the LED cryptosystem has been done, and no redundant model of LED has been proposed. During laser fault injection, it is possible to introduce redundant faults. We expect to recover the secret key without being affected by redundant faults. First, we use the impossible differential fault path to classify the location of the faults; afterwards, we use the inverse algebraic differential equations to estimate the subkey. In 1300 experiments, 100% of subkeys could be recovered by only 6 attacks, and 92% subkey could be recovered in no more than 8 minutes. Compared to the existing attacks, the number of attacks and the needed time is significantly reduced. Our results have verified that the LED-64 cryptosystem is significantly vulnerable to the IDFAA. We also expect that IDFAA and the RRNFM will help researchers further improve the block cipher security.

ACKNOWLEDGMENT

(Liang Dong and Hongxin Zhang are co-first authors.)

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Authors' photographs and biographies not available at the time of publication.

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